Every condition decides an initial segment of the jump by a finite extension

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Editorial note. The counted unit is the article's own Lemma lem:coh-wf-forcing-jump at source lines 507-509 together with the proof environment that begins on the next line (510-520); the pairing is the ordinary adjacent one.

Editorial note. The retained context at source lines 480-491 is the author's own account of conditions, compatibility and the two forcing clauses for the jump formulas; the retained context at source line 505 is his own explanation of the relation forcing a decided initial segment of the jump.

Editorial note. The retained preamble reproduces verbatim the author's document class at source line 2, his theorem declarations at source lines 16-25, his macro definitions at source lines 46-88 and 91, his natural-operation macros at source lines 104-107 and his quotation macro at source line 116. The LS2 font-encoding block at source lines 93-99, which declares \smashtimes through the stix fonts, is not retained because the host TeX Live has no stix directory and because the author never uses that symbol: source line 105 defines \ntimes as \dottimes and source line 110 shows the alternative already commented out. mathabx (source line 103) is retained because \dotplus and \dottimes come from it. The cleveref load at source line 6 and the inputenc load with the utf8x option at source line 8 are not retained, for the assembler's own cleveref machinery and the forced xelatex engine respectively. Nothing was renamed, redefined or added.

Editorial note. No retained line contains a citation or a cross-reference, so the material carries no bibliography and needs no navigation pointer.

Editorial note. The article's conservation theorems, its lemma lem:shore-blocking-wf, its lemma lem:wf-product, its lemma lem:coh-wf-forcing-wf and its proposition prop:coh-wf are not counted here and are not used as premises.

Allowed standard prerequisites (editorial).

- Conditions of the Cohen forcing used for COH and the compatibility relation. A condition is a pair (sigma, rho) with sigma a finite binary string and rho an M-finite approximation of the sequence of sets to be made cohesive, readable as a Mathias condition; a string tau is compatible with

(sigma, rho) when sigma is an initial segment of tau and the difference tau minus sigma is contained in the members of the approximation; the article's own account is retained at source lines 480-483.. Fixes the condition (sigma,rho) of the counted lemma and the compatibility relation used both to define the set W and to transfer the forcing statement to the generic G.

- The forcing relation for the existential and universal jump formulas. A condition forces the existential statement that some value of the functional on the generics is defined when a witness and a bound smaller than the length of sigma exist, and it forces the universal statement that all values diverge when every compatible string fails to converge anywhere; the article's own definition is retained at source lines 485-491, and its explanation of forcing a decided initial segment of the jump at source line 505.. The counted lemma concludes a statement of the form that a condition forces the k-th initial segment of the jump to equal a fixed string, whose meaning is exactly the retained relation at source line 505.

- Sigma^0_1 definability of the set of candidates and M-finiteness. The set of strings that admit a compatible extension deciding a given pattern is described by a Sigma^0_1(Y) formula, hence is M-finite in a model of RCA_0; the article uses this M-finiteness repeatedly in this section.. The proof forms the set W of candidate strings, argues that it is Sigma^0_1(Y) and M-finite, and then takes a lexicographically maximal element, which needs M-finiteness.

One can think of a condition (σ, ρ) as a Mathias condition $(\sigma, \vec{R}_\rho)$. Requiring that $\rho \prec Q$ ensures that $\vec{R}_\rho$ is M-infinite. When increasing the length of ρ, we make progress in cohesiveness. Note that being a condition is a $\Delta^0_1(Q)$ predicate, as it simply requires checking that $\sigma \prec Q$, while the extension relation is $\Delta^0_1(Y)$. In particular, both are $\Delta^0_1(Q \oplus Y)$, so $\Delta^0_1(P)$.

A sequence τ is said to be *compatible* with a condition (σ, ρ) if $\sigma \prec \tau$ and $\tau - \sigma \subseteq \vec{R}_\rho$.

Definition 0.1. *Let (σ, ρ) be a condition, and Φ_e be a Turing functional.*

1. $(\sigma, \rho) \Vdash \exists x \Phi_e^{G \oplus Y}(x) \downarrow$ *if there is some* $x, t < |\sigma|$ *such that* $\Phi_e^{\sigma \oplus Y}(x)[t] \downarrow$
2. $(\sigma, \rho) \Vdash \forall x \Phi_e^{G \oplus Y}(x) \uparrow$ *if for every* τ *compatible with* (σ, ρ) *and every* $x \in M$, $\Phi_e^{\tau \oplus Y}(x) \uparrow$.

We will need the following two technical lemmas, which are Shore blocking arguments for satisfying the two kind of requirements, namely, deciding the jump and preserving $\mathsf{WF}(\alpha)$. In what follows, $(\tau, \rho) \Vdash (G \oplus Y)' \upharpoonright_k = \sigma'$ means that for every $e < |\sigma'|$, if $\sigma'(e) = 1$ then $(\tau, \rho) \Vdash \Phi_e^{G \oplus Y}(e) \downarrow$ and if $\sigma'(e) = 0$, then $(\tau, \rho) \Vdash \Phi_e^{G \oplus Y}(e) \uparrow$. Thus, this relation is $\Pi^0_1(Y)$.

Lemma 0.2. *lemma/lem:coh-wf-forcing-jump Let (σ, ρ) be a condition. For every $k \in M$, there exists some M-finite $\sigma' \in 2^k$ and an extension $(\tau, \rho) \leqslant (\sigma, \rho)$ such that $(\tau, \rho) \Vdash (G \oplus Y)' \upharpoonright_k = \sigma'$.*

Proof. Let W be the set of all $\sigma' \in 2^k$ such that $(\exists \tau \succ \sigma)(\exists t)(\forall e < k)(\tau - \sigma \in \vec{R}_\rho \wedge |\tau - \sigma| \geqslant 1 \wedge \sigma'(e) = 1 \to \Phi_e^{\sigma \oplus Y}(e)[t] \downarrow)$. Note that W is $\Sigma^0_1(Y)$, hence is M-finite. Moreover, W is non-empty, as it contains the string $000\ldots$. Let $\sigma' \in W$ be the lexicographically maximal element, and let $\tau \succeq \sigma$ witness that $\sigma' \in W$.

We claim that (τ, ρ) forces $(G \oplus Y)' \upharpoonright_k = \sigma'$. Fix some $e < k$.

Case (a): $\sigma'(e) = 1$. Then $\Phi_e^{\tau \oplus Y}(e) \downarrow$, hence for every G compatible with (τ, ρ), $e \in (G \oplus Y)'$.

Case (b): $\sigma'(e) = 0$. The maximality of σ' ensures that for every μ compatible with (τ, ρ), $\Phi^{\mu \oplus Y}(e) \uparrow$. It follows that $\Phi_e^{G \oplus Y}(e) \uparrow$, hence $e \notin (G \oplus Y)'$. $\square$